

Zero-Record Certification of Reversible Quantum Histories

Causal memory, predicate readout, recoherence, and the limits of endpoint evidence

Zero-Record Quantum Histories Project

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Abstract

We study whether a controlled quantum process can contain a nontrivial internal history, including memory and conditional action, while retaining only a selected predicate as an interferometric phase after all branch-distinguishing records are uncomputed. The elementary circuit is physically valid but not, by itself, evidence for an internal history: a known closed sequence $U_H^\dagger O_p U_H$ is endpoint-equivalent to a directly compiled effective unitary. We make this limitation central. We define a zero-record condition as the absence of residual information that distinguishes histories within each allowed predicate class, formulate reset, leakage, visibility, completeness, and soundness criteria, and separate closed-circuit validation from causal certification under restricted temporal access. We prove phase survival, endpoint compression, and no-transcription statements; show why reset fidelity alone cannot certify recoherence; and formulate an analytic late-challenge reversible-memory benchmark. A density-matrix reference implementation reproduces the exact visibility–environment-overlap relation, compares coherent, incoherent, retained-memory, and direct-phase controls, and quantifies dephasing, amplitude damping, depolarisation, environmental leakage, inversion error, depth scaling, and coarse-grained marginal transcript leakage. The results establish a reproducible baseline and a set of kill criteria. They do not establish a device-independent causal witness, an experimental late-challenge score, cryptographic zero knowledge, a quantum advantage, objective collapse, or a preferred interpretation of quantum mechanics. The open scientific target is a resource-bounded separation that certifies causal memory while preserving zero-record recoherence.

Keywords: reversible quantum computation; multitime quantum processes; causal memory; interferometry; uncomputation; quantum records; recoherence.

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1 Introduction

Quantum theory permits a controlled system to correlate distinguishable alternatives with internal records and later erase those records by reversing the responsible dynamics. If a Boolean function of an internal record is transferred to a relative phase before reversal, the auxiliary systems can return to their reference states while the phase remains observable. This combination of coherent control, phase kickback, uncomputation, and interference is standard quantum information processing [1, 2]. It is nevertheless a useful starting point for a sharper question: how much *causal history* can occur inside such an evolution, and what evidence for that history can survive if its transcript is physically removed?

The question contains an immediate identifiability problem. Suppose a known laboratory circuit creates a record, calculates a predicate, and exactly reverses the record. The induced input–output channel can be compiled into a direct unitary that never instantiates the advertised memory. No measurement confined to the endpoints distinguishes two implementations of the same channel. Describing intermediate components as histories, observers, or branches cannot alter this fact. Any claim of a genuine internal history therefore needs additional structure: intermediate intervention ports, delayed inputs, locality constraints, black-box access, a memory-dimension witness, or a resource separation.

This paper develops a conservative baseline around that limitation. We use the term *reversible quantum history* for a sequence of explicitly modelled transformations and correlations, not for an ontologically unique world. A *zero-record* protocol retains an allowed predicate while removing residual information that distinguishes histories sharing that predicate. The condition is stronger than resetting one named memory register, because uncontrolled environments and computational garbage can preserve a transcript. It is weaker and conceptually different from cryptographic zero knowledge: without a simulator-based adversarial definition, zero-record is a physical decoupling criterion, not a cryptographic security claim.

Our contributions are fourfold. First, we provide a formal separation between a closed-circuit demonstration and a causal-certification task. Second, we collect exact consequences of unitary quantum mechanics that constrain any proposal: phase survival under uncomputation, endpoint compression, the record–visibility relation, and the impossibility of retaining a perfectly distinguishing transcript while recovering perfect interference. Third, we define a family of benchmark instances, including a simulated reversible finite-state transducer and a separate analytic late-challenge game. Fourth, we supply an independently reproducible density-matrix simulator, controls, noise sweeps, tests, data, and figures.

The intended outcome is not a spectacular interpretation-dependent claim. It is an audit-friendly research object that makes clear which results are known, which are derived here, which are numerical, and which remain conjectural. In particular, this version does not prove that the joint primitive is novel in the priority sense, does not produce a device-independent witness of an internal agent, and does not show a coherent-over-incoherent complexity separation. Those are the next decisive units of progress.

1.1 Operational question

The target question is:

Can a verifier certify, relative to explicit access and resource assumptions, that a quantum process used internal causal memory, learn only a selected coarse-grained predicate of the resulting history, and finish with both negligible within-predicate transcript leakage and recoverable interference?

The clauses are deliberately conjunctive. Interference without causal certification is ordinary coherent control. A causal-memory witness without reversal is a temporal-correlation

experiment. Uncomputation without a nontrivial access model is a compiler transformation. Predicate privacy without a non-injective coarse-graining can be vacuous. The proposed research programme begins only at their intersection.

2 Scope, evidence levels, and adjacent literature

2.1 Epistemic firewall

We distinguish four levels of statement throughout:

1. **Operational:** specified coherent evolutions yield an interference signal and controls do not.
2. **Model-relative historical:** components of a declared decomposition carry different internal records at intermediate times.
3. **Interpretive:** one may describe those components using a preferred interpretation of quantum mechanics.
4. **Unsupported:** the experiment accesses a pre-existing macroscopic alternative world or establishes a unique ontology.

Only the first level is directly measured. The second follows from a validated model and declared tensor decomposition. The third is underdetermined by shared unitary predictions. The fourth is outside the protocol.

2.2 Known ingredients

Relative-state language and decoherence motivate discussions of correlated components and records [3, 4]. Consistent-histories methods describe sequences and their decoherence functionals [5, 6], although alternatives that are deliberately recombined are not simultaneously a classical consistent family. Reversible computation supplies compute–copy-or-phase–uncompute constructions [1]. Wave–particle duality and quantum erasure quantify the loss and conditional recovery of interference when path information exists [7, 8]. Measurement reversal and error-correction demonstrations establish that selected measurement interactions can be undone under controlled conditions [9, 10].

For multitime structure, quantum combs and process tensors treat an experiment as a process with intervention slots rather than as a single endpoint channel [11, 12, 13]. Quantum causal discovery makes interventions central to the distinction between correlation and causation [14]. Temporal studies can diagnose instrument-specific memory effects, bound effective environment dimension, and, under stronger assumptions, certify a quantum memory without trusting its implementation [15, 16, 17]. A January 2026 preprint reports a trapped-ion proof of principle for device-independent two-time memory certification [18]; its peer-review status at our cutoff is provisional.

Wigner-friend-inspired protocols show how a small physical record can play the functional role of an internal observer, including explicit analyses of memory erasure [19, 20, 21, 22]. These works do not place a conscious human observer in coherent superposition. The memory-erasure result is a particularly direct antecedent and prevents a novelty claim based merely on a friend acquiring and then losing a record.

Quantum zero-knowledge proof systems establish rigorous ways for a verifier to learn validity without learning a witness, including QMA, classical-verifier protocols for quantum computations, non-destructive state proofs, and superposition-secure transcript models [23, 24, 25, 26]. Their security definitions are computational or information-theoretic statements about simulators and adversaries. Our zero-record criterion instead asks whether physical residual systems remain correlated with distinctions inside a predicate class. The analogy is generative but not an equivalence.

Quantum algorithmic measurements (QUALMs) provide a complexity-theoretic language for experimental access and exhibit tasks with exponential separations between coherent and incoherent access [27]. Computational mechanics supplies irreducible predictive-memory measures, and quantum models or adaptive agents can reduce memory costs for selected stochastic processes [28, 29]. These results suggest the right form of a future originality claim: not that a circuit used superposition, but that a precisely defined historical-certification task has a provable resource separation.

2.3 Evidence and novelty discipline

We classify claims as known results, derivations, numerical observations, conjectures, or open objectives. The accompanying literature review is a scoping review rather than a PRISMA systematic review. Several directly adjacent manuscripts appeared as 2026 preprints and remain provisional at the cutoff date. Consequently, the phrase “not found in this search” is never promoted to “never proposed.” A priority claim requires database-complete searches, forward and backward citation tracing, and specialist review.

The elementary history–predicate–uncompute circuit is not new. Generic memory–visibility inequalities are also heavily constrained by existing complementarity and channel-recovery results [7, 30]. The candidate open intersection is the joint task of causal-memory certification, within-predicate zero-record leakage, and subsequent recoherence under an explicit access model.

2.4 Adversarial novelty map

Table 1 records which intuitive headlines are already occupied. It is deliberately stricter than a conventional motivation section.

Table 1: Adjacency audit at the 12 August 2026 cutoff. The final row is a candidate research task, not an established priority claim.

Candidate headline	Existing territory	Remaining requirement
Compute a predicate and erase work	Reversible computation and phase kickback	No novelty without a new access or resource theorem
Erase an internal observer’s memory	Wigner-friend memory erasure is explicit prior work	Certify causal memory and nonvacuous final privacy
Bound memory by temporal data	Process-tensor and memory-dimension witnesses exist	Combine the witness with a later coherent inverse
Hide a quantum witness or transcript	Quantum zero-knowledge protocols cover strong adversarial models	Relate simulator security to physical final-record decoupling
Show coherent experimental advantage	QUALMs already give exponential separations for other tasks	Prove a separation for the zero-record task itself
Trade objectivity against recovery	Darwinism, encoding transitions, and QEC tradeoffs are active	Only a narrower operational theorem may remain
Causal memory + predicate-only disclosure + final decoupling + recoherence	No single primary source located by the bounded search	Formal anti-shortcut model and nontrivial theorem still required

2.5 Direct 2026 priority risks

The current literature makes interpretation-led framing especially risky. Three January 2026 preprints use inter-branch communication language, compile Wigner-friend-style message circuits, or benchmark them on superconducting processors [31, 32, 33]. Another January preprint studies approximate records, decoherence, recoherence, and recovery in long isolated histories [34]. These manuscripts are not treated as settled peer-reviewed results, but they are direct priority evidence against presenting branch communication, memory erasure, or generic recoherence as new.

Quantum Darwinism formalises redundant proliferation of classical records, and models already connect redundant objective information with consistent histories and quantum encoding [35, 36, 37]. A March 2026 peer-reviewed model further distinguishes approximate decoherent histories from the redundant records associated with objectivity [38]. A preprint posted eight days before our cutoff claims an exact QEC–Darwinism tradeoff and a model-independent no-go statement [39]. Its generality requires independent scrutiny, but its existence makes objectivity versus recoverability a poor novelty headline.

Thermodynamic erasure is also not an empty intersection. Complexity-constrained quantum thermodynamics derives work–complexity tradeoffs for reset [40], while microscopic measurement models study records, reset, finite-time control, and work [41]. Our inverse is logically reversible; assigning $k_B T \ln 2$ to every uncomputed bit would conflate uncomputation with blind erasure.

2.6 Macroscopic scale and nonunitarity

Matter-wave experiments have extended coherent interference to increasingly massive molecules and nanoparticles, including a 2026 sodium-nanoparticle result [42]. This calibrates centre-of-mass coherence, not reversible control of a macroscopic internal observer and its environment. Objective-collapse models make parameterised predictions and are experimentally constrained, including by a March 2026 XENONnT spontaneous-radiation search [43, 44, 45, 46]. An unexplained visibility loss in the present protocol would first indicate an incomplete noise or control model; it would not by itself identify fundamental nonunitarity.

2.7 Conservative novelty statement

The dated search found primary literature for every clause separately but did not locate a protocol that jointly imposes: (i) an interventional lower bound on causal memory, (ii) disclosure limited to a non-injective predicate, (iii) final decoupling of every other accessible history record, (iv) recovered interference, and (v) an explicit shortcut class. This is a bounded negative search result, not proof of absence. A first submission must add citation-network traversal, subscription-database searches, patent and thesis checks, and independent specialist review before using a phrase such as *to our knowledge*.

3 Model and operational quantities

3.1 Registers and histories

Let

$$\mathcal{H} = \mathcal{H}_B \otimes \mathcal{H}_W \otimes \mathcal{H}_M \otimes \mathcal{H}_G \otimes \mathcal{H}_A \otimes \mathcal{H}_E. \quad (1)$$

The label register B coherently indexes alternatives; W is a controlled world; M is internal memory; G contains reversible work registers; A is the phase-readout ancilla; and E comprises

degrees of freedom that are not inverted. We abbreviate the intended-reset history registers as $R = WMG$. Register A is tracked separately because it is prepared and returned in $|-\rangle$ rather than $|0\rangle$.

A T -step history of label b is a sequence

$$\mathbf{U}_b = (U_{b,T}, \dots, U_{b,1}) \quad (2)$$

with declared intervention slots between steps. In a closed implementation the controlled history unitary is

$$U_H = \sum_b |b\rangle\langle b|_B \otimes U_b, \quad U_b = U_{b,T} \cdots U_{b,1}. \quad (3)$$

The word history denotes this model-relative temporal structure. It is not assumed to be a preferred-basis ontological primitive.

The initial state is

$$|\Psi_0\rangle = \sum_b \alpha_b |b\rangle_B |0\rangle_R |-\rangle_A |e_0\rangle_E, \quad \sum_b |\alpha_b|^2 = 1. \quad (4)$$

An ideal history stage yields

$$|\Psi_H\rangle = \sum_b \alpha_b |b\rangle_B |r_b\rangle_R |-\rangle_A |e_b\rangle_E. \quad (5)$$

The environment states need not be orthogonal. Their distinguishability quantifies uncontrolled which-history information.

3.2 Predicate readout

Let p be a function on the modelled internal history. A clean predicate oracle acts on the populated history subspace as

$$O_p |b\rangle |r_b\rangle = e^{i\phi(p_b)} |b\rangle |r_b\rangle, \quad (6)$$

with the binary choice $\phi(p) = \pi p$ used in the reference protocols. This action can be constructed by reversibly computing p into a work bit, applying phase kickback from a $|-\rangle$ ancilla, and uncomputing the work bit. Calling an arbitrary phase gate a predicate oracle would hide the very implementation that the project seeks to analyse, so the simulator audits all work registers.

3.3 Visibility and distinguishability

For two balanced alternatives, the reduced control state can be written

$$\rho_B = \frac{1}{2} \begin{pmatrix} 1 & c \\ c^* & 1 \end{pmatrix}. \quad (7)$$

We define visibility $V = 2|\rho_{B,01}| = |c|$. If all controlled systems are reset and only pure environmental records remain, $c = e^{i\Delta\phi} \langle e_1 | e_0 \rangle$. The optimal pure-state record distinguishability is

$$D = \frac{1}{2} \| |e_0\rangle\langle e_0| - |e_1\rangle\langle e_1| \|_1 = \sqrt{1 - V^2}. \quad (8)$$

The simulator also reports l_1 coherence for more than two histories, final control purity, and the probability of the expected interferometric outcome. Visibility is not inferred from reset fidelity.

3.4 Reset and leakage

For a target controlled state $|0\rangle_R$,

$$F_{\text{reset}} = \langle 0 | \rho_R^f | 0 \rangle. \quad (9)$$

This quantity checks the declared controlled registers only. A history-dependent E can coexist with $F_{\text{reset}} = 1$ and $V = 0$.

To define transcript leakage, introduce a classical reference H for fine-grained history labels and an allowed predicate $P = p(H)$. Let T collect all accessible residual transcript systems, including R and any modelled accessible part of E . For conditional residual states ρ_T^h and a declared distribution q_h , form the cq state

$$\omega_{HPT} = \sum_h q_h |h\rangle\langle h|_H \otimes |p(h)\rangle\langle p(h)|_P \otimes \rho_T^h. \quad (10)$$

We define

$$L_{\text{ZR}} = I(H : T | P)_\omega. \quad (11)$$

Equivalently, it is the average within-predicate Holevo information for this cq construction. It asks whether the residual distinguishes microhistories that the permitted output deliberately coarse-grains. Because it is distribution-weighted, a small value is not a worst-case guarantee unless every fibre is bounded separately. Moreover, L_{ZR} does not constrain records of P itself; the separate visibility clause is needed to rule out a predicate record that destroys coherence between predicate classes.

Remark 3.1 (Non-injective predicates). If p is injective on the support of H , then H is already determined by P and $I(H : R | P) = 0$ for every residual record. The condition is vacuous. A meaningful zero-record benchmark therefore needs at least two histories in one predicate fibre, as in the four-history reference instance.

3.5 Causal access model

A multitime process is represented by a process tensor $\Upsilon_{T;0}$ whose slots accept instruments or coherent controls. Let $\mathfrak{C}_{d,L}$ denote an explicitly declared comparison class, for example processes with memory dimension at most d , causal depth at most L , no access from the response module to B , or no adaptive dependence on a late challenge. A causal witness is meaningful only relative to such a class.

Definition 3.2 (Zero-record causal certification). *Fix a declared history distribution whose predicate is non-injective on its support. A protocol family has parameters $(c, s, \varepsilon, \eta, v; \mathfrak{A}, \mathfrak{C})$ if valid processes in $\mathfrak{A}$ are accepted with probability at least c , comparison processes in $\mathfrak{C}$ are accepted with probability at most $s < c$, $L_{\text{ZR}} \leq \varepsilon$, $F_{\text{reset}} \geq 1 - \eta$, and $V \geq v$.*

The definition is a target specification. The reference implementation establishes reset, visibility, and selected marginal leakage diagnostics for trusted circuit families. Except in cases where a common final state or the full joint record is known analytically, separate marginal leakage values do not certify joint transcript decoupling. The implementation does not establish a general device-independent completeness–soundness gap.

4 Structural results and no-go statements

This section separates exact statements from the open certification problem. The first results are elementary consequences of unitarity, but writing them together prevents a physically correct demonstration from being misreported as evidence for a uniquely realised internal narrative.

4.1 Phase survival under uncomputation

Theorem 4.1 (Phase survival). *Let U_H satisfy $U_H|b\rangle|0\rangle = |b\rangle|r_b\rangle$ on a control basis, and let a clean oracle satisfy*

$$O_p|b\rangle|r_b\rangle = e^{i\phi_b}|b\rangle|r_b\rangle. \quad (12)$$

Then, for every superposition $\sum_b \alpha_b|b\rangle$,

$$U_H^\dagger O_p U_H \sum_b \alpha_b|b\rangle|0\rangle = \sum_b \alpha_b e^{i\phi_b}|b\rangle|0\rangle. \quad (13)$$

All registers included in the inverse return to their reference state while the relative phase remains on the control.

The result requires a clean oracle. If the predicate computation leaves states $|g_b\rangle$ outside the inverse, the control coherence is multiplied by their overlaps. Copying the predicate into a classical record and retaining it is therefore not innocuous.

4.2 Endpoint compression

Theorem 4.2 (Endpoint indistinguishability). *Let two implementations induce the same completely positive trace-preserving map $\mathcal{E}$ from every admissible input to every accessible output. No experiment restricted to those endpoints distinguishes the implementations. In particular, if a known closed history circuit resets R exactly and induces V_B on B , a direct implementation of V_B is operationally identical at the endpoints.*

Proof. For every input ρ and output POVM element E_y , both implementations yield $\text{Tr}[E_y \mathcal{E}(\rho)]$. Exact equality of channels implies zero diamond distance and hence zero discrimination advantage. When R is returned to a fixed pure state by a global unitary, preservation of inner products makes the induced map on the supported subspace of B an isometry, and therefore a unitary when input and output dimensions agree. A direct circuit for that unitary has the same endpoint channel. $\square$

For the binary predicate circuit, $V_B = \sum_b e^{i\phi_b}|b\rangle\langle b|$. This theorem is the reason the direct-phase control must reproduce the ideal endpoint statistics. A discrepancy would diagnose an implementation error, not historical authenticity.

Corollary 4.3 (No endpoint-only agent witness). *The existence, memory size, policy depth, or phenomenological interpretation of an intermediate subsystem cannot be inferred from endpoint data alone when a comparison implementation realises the same channel.*

4.3 No-transcription and visibility

Theorem 4.4 (Pure-record complementarity). *After all intended controlled inverses, suppose two balanced alternatives have the pure global state*

$$|\Psi\rangle = \frac{|0\rangle|r_0\rangle + e^{i\phi}|1\rangle|r_1\rangle}{\sqrt{2}}. \quad (14)$$

Then

$$V = |\langle r_1|r_0\rangle|, \quad D = \frac{1}{2} \|\langle r_0|r_0\rangle - \langle r_1|r_1\rangle\|_1 = \sqrt{1 - V^2}. \quad (15)$$

Thus an orthogonal transcript gives $V = 0$, while $V = 1$ implies that the two record states differ at most by a global phase and are not distinguishable.

Proof. Tracing out the record leaves off-diagonal control element $e^{-i\phi}\langle r_1|r_0\rangle/2$, giving the visibility. The trace distance of two pure states has the stated closed form. $\square$

For mixed records σ_0, σ_1 , define root fidelity

$$F(\sigma_0, \sigma_1) = \|\sqrt{\sigma_0}\sqrt{\sigma_1}\|_1. \quad (16)$$

Uhlmann's theorem and the Fuchs–van de Graaf inequalities [47, 48] imply

$$V \leq F(\sigma_0, \sigma_1), \quad D(\sigma_0, \sigma_1)^2 + V^2 \leq 1 \quad (17)$$

for the corresponding two-path experiment. The first inequality may be strict because uncontrolled phases or additional correlations can reduce actual coherence below the maximum compatible with the reduced records.

Proposition 4.5 (Reset is not recoherence). *Perfect reset of the declared controlled register does not imply nonzero visibility.*

Proof. Consider

$$|\Psi\rangle = \frac{|0\rangle_B|0\rangle_R|e_0\rangle_E + |1\rangle_B|0\rangle_R|e_1\rangle_E}{\sqrt{2}} \quad (18)$$

with $\langle e_1|e_0\rangle = 0$. The controlled register has $F_{\text{reset}} = 1$, but tracing out E gives a fully dephased B and $V = 0$. $\square$

This counterexample requires reporting reset and leakage separately. A high local reset fidelity can coexist with a complete transcript elsewhere.

4.4 Exact meaning of zero record

Proposition 4.6 (Within-predicate decoupling). *For the cq state in the definition of L_{ZR} ,*

$$L_{\text{ZR}} = \sum_p q_p \chi\left(\{q(h|p), \rho_T^h\}_{h:p(h)=p}\right). \quad (19)$$

It vanishes if and only if all residual states with nonzero weight inside each predicate fibre are identical.

Proof. Conditioning on the classical value $P = p$ gives a cq mutual information equal to the Holevo information of that fibre. Holevo information is a weighted sum of relative entropies to the average state and is zero exactly when every supported conditional state equals that average. $\square$

Exact common-state decoupling is sufficient but not necessary: a residual may depend on the allowed predicate while carrying no extra within-class transcript. Such predicate dependence can nevertheless suppress interference between predicate classes, which is why zero-record leakage and visibility are separate requirements. For the clean interferometric construction used here, the stronger target is $\rho_R^h = |0\rangle\langle 0|$ for every h together with a history-independent state of every unresolved transcript system.

4.5 A late-challenge memory bound

The following classical bound supplies a transparent causal comparison for the trusted reference protocol. It is not a device-independent quantum-memory theorem.

Proposition 4.7 (Two-bit late-challenge game). *Let $H = (H_0, H_1)$ be uniform. Before an independent uniform challenge $X \in \{0, 1\}$ is chosen, an encoder maps H to a classical memory*

M with at most d distinguishable states. After H becomes inaccessible, a response module sees only (M, X) and returns Y . Shared or private randomness is allowed. Then

$$d = 1 : \quad \Pr[Y = H_X] \leq \frac{1}{2}, \quad (20)$$

$$d = 2 : \quad \Pr[Y = H_X] \leq \frac{3}{4}, \quad (21)$$

$$d = 4 : \quad \Pr[Y = H_X] = 1 \quad \text{is achievable.} \quad (22)$$

Proof. The success probability is affine in the encoding and decoding strategies, so shared or private randomisation cannot outperform a deterministic extreme point. With one memory state, Y is independent of both unbiased target bits. With two memory states the encoder transmits one classical bit. For each message, the optimal decoder returns the majority value of each requested coordinate among strings mapped to that message. Enumerating the partitions of four two-bit strings, or applying the standard $2 \rightarrow 1$ random-access-code bound, gives at most six correct cases among the eight equally likely (H, X) pairs. Storing both bits in four states achieves all eight. $\square$

A qubit memory can exceed the one-classical-bit value using a quantum random access code [49], so the comparison class must say whether memory dimension is classical or quantum. A classical two-bit memory reaches unit success in this readout game. The coherent phase construction in the next section uses the same response function $h \mapsto h_X$ but implements it as a phase oracle over all h ; its endpoint measurement is not a sampled value of H_X . Consequently, the random-access-code probabilities above are not by themselves a soundness bound for the coherent phase task. The current density-matrix simulator also does not implement an independent random challenge register or estimate this game score. Turning the proposition into a certification test remains open and requires the stated access assumption that the response module cannot read B directly.

4.6 What is and is not proved

The exact results imply that predicate survival and transcript erasure are compatible in unitary quantum mechanics. They also prove that endpoint evidence cannot establish the advertised intermediate implementation. The late-challenge game supplies a model-relative memory test, but the present work does not derive a single-shot joint inequality whose violation simultaneously certifies quantum memory, zero-record decoupling, and recoherence. We state the desired separation only as a conjecture.

Conjecture 4.8 (Resource-bounded historical separation). *There exists a family of black-box multitime tasks for which coherent access with reversible internal memory achieves zero-record predicate readout using polynomial resources, whereas every protocol in a declared incoherent or bounded-memory comparison class requires superpolynomial resources or has a nonzero soundness gap.*

Establishing or refuting this conjecture is the main theoretical criterion for continuing the programme beyond its benchmark phase.

5 Reference protocol family

The reference family is deliberately small. ZR-0, ZR-1, ZR-2, and ZR-4 motivate components of the exact density-matrix simulator, while ZR-3 is an analytic late-challenge task not scored by the present code. Their purpose is to expose every record and control, not to maximise qubit count. Table 2 summarises the instances.

Table 2: Reference instances and the question each one tests.

Instance	Minimum structure	Diagnostic purpose
ZR-0	B, M, A	Exact phase survival and endpoint direct-phase equivalence
ZR-1	B, M, A, E	Visibility versus pure environmental-record overlap
ZR-2	multistep W, M, G	Reversible observation, state update, action, and reset
ZR-3	two-bit B, M , late X , response Y	Analytic model-relative causal-memory challenge
ZR-4	four histories, non-injective p	Nonvacuous within-predicate transcript leakage

5.1 ZR-0: minimal phase survival

Prepare

$$|+\rangle_B |0\rangle_M |-\rangle_A. \quad (23)$$

A CNOT from B to M creates the record

$$\frac{|0\rangle_B |0\rangle_M + |1\rangle_B |1\rangle_M}{\sqrt{2}} |-\rangle_A. \quad (24)$$

A CNOT from M to the phase ancilla contributes $(-1)^m$. Reapplying the first CNOT uncomputes the memory and leaves

$$|-\rangle_B |0\rangle_M |-\rangle_A. \quad (25)$$

A final Hadamard on B returns outcome one with unit probability. Applying Z_B directly produces the same endpoint state. Omitting the predicate produces outcome zero, while retaining the memory makes the reduced control incoherent and its Hadamard-basis outcome uniform.

This three-register circuit tests bookkeeping. It is not a historical-certification claim.

5.2 ZR-1: tunable environmental record

After the intended controlled reset, let the two alternatives correlate with

$$|e_0\rangle = |0\rangle, \quad |e_1\rangle = \gamma|0\rangle + \sqrt{1-\gamma^2}|1\rangle, \quad 0 \leq \gamma \leq 1. \quad (26)$$

The reduced control coherence is multiplied by γ , giving the analytic target

$$V = \gamma, \text{quad}D = \sqrt{1-\gamma^2}. \quad (27)$$

This two-path construction gives the complementarity circle analytically. The four-history simulator instead sweeps a constant pairwise overlap η , for which its normalised multipath l_1 coherence equals η ; it does not identify that multipath quantity with the two-path distinguishability D . At zero overlap an orthogonal external record survives even though the named memory has reset perfectly. At unit overlap the environment contains no distinguishable path record.

5.3 ZR-2: reversible finite-state agent

An agent is modelled functionally as a reversible transducer rather than as a conscious observer. At each time t it:

1. receives a world symbol w_t ;
2. updates a finite internal state m_t ;
3. writes a reversible action a_t ;
4. preserves enough work information to invert the update.

The implemented transducer stores the two-bit world symbol in memory. In each round it computes a challenge-dependent decision into a work bit, conditionally flips one world bit, and uncomputes the work bit. The final internal predicate combines world parity with the recomputed decision parity; on the intended path this equals label parity. Deleting an action gate while keeping that oracle fixed changes the eventual predicate. Each update is a permutation of computational basis states, hence a unitary. The inverse applies action, decision, memory, and world gates in reverse order. Unit tests verify the permutation and inverse identities on every basis state, not merely on the chosen input.

The point of ZR-2 is to enforce temporal and causal structure in the implementation. By the endpoint theorem, the trusted closed circuit remains compressible to its effective phase oracle.

5.4 ZR-3: analytic late-challenge random-access history

Let the fine-grained history be $h = (h_0, h_1) \in \{0, 1\}^2$. Prepare a coherent uniform label and copy both bits into a two-qubit memory during the observation window:

$$\frac{1}{2} \sum_{h_0, h_1} |h_0 h_1\rangle_B |h_0 h_1\rangle_M. \quad (28)$$

Only after this correlation exists, choose a classical challenge $x \in \{0, 1\}$. A response module that cannot access B computes $y = m_x$ into a response qubit, phase-marks y , and uncomputes the response. The inverse observation then returns M to $|00\rangle$. The final control state is

$$Z_{B_x} |++\rangle_B, \quad (29)$$

up to a known convention, so an X -basis measurement identifies the selected phase pattern.

In the corresponding *classical readout game*, across random challenges a memoryless response succeeds at most one half of the time, one classical bit at most three quarters, and two bits perfectly. A quantum one-bit comparison has a different bound and must not be grouped with the classical class. The coherent circuit above instead applies the phase function $h \mapsto h_x$ to every history amplitude. Its final X -basis measurement diagnoses the phase pattern; it is not a draw of the classical answer H_X . Thus the classical success bounds do not directly certify the coherent endpoint channel.

The timing choice blocks a classical response prepared before x from knowing which stored feature will be queried. It does not block a malicious device with unauthorised access to B from applying Z_{B_x} directly. Thus ZR-3 is a proposed causal benchmark under an architectural access assumption, not a device-independent proof. The current simulator's challenge tuple consists of fixed gate parameters applied after memory formation in circuit order; it contains no independent X register and does not report the ZR-3 game score.

5.5 ZR-4: a nontrivial privacy fibre

Take the same four histories and the allowed predicate

$$p(h) = h_0 \oplus h_1. \quad (30)$$

Each predicate value contains two fine histories. The branch-conditioned residual states define a meaningful conditional Holevo information. In the exact protocol all controlled residual states equal $|0\rangle$, so $L_{\text{ZR}} = 0$. A retained copy of h gives one bit of additional information within each parity class, while partial nonorthogonal records interpolate continuously.

This instance prevents a common definitional error: with only two histories and predicate $p(h) = h$, conditioning on the allowed predicate already reveals the complete history and makes conditional leakage zero regardless of the residual transcript.

5.6 Control matrix

Table 3: Predicted qualitative outcomes for the mandatory controls. “Reset” refers only to declared controlled records; E is assessed separately.

Condition	Phase response	V	Reset	Interpretation
Ideal coherent	yes	1	1	Positive implementation control
Direct compiled phase	yes	1	1	Endpoint-equivalent shortcut
No predicate	no	1	1	Negative predicate control
Classical label mixture	no relative phase	0	not diagnostic	Incoherent baseline
Memory retained	local phase inaccessible	0	0	Which-history record remains
Orthogonal E record	local phase inaccessible	0	1	Reset does not imply recoherence
Perturbed inverse	degraded	< 1	< 1	Ordinary control error
Memoryless late response	bounded	varies	1	Causal comparison class

6 Reference simulations

6.1 Implementation

The simulator evolves an explicit 256×256 density matrix on

$$\mathcal{H}_B(4) \otimes \mathcal{H}_W(4) \otimes \mathcal{H}_M(4) \otimes \mathcal{H}_G(2) \otimes \mathcal{H}_A(2). \quad (31)$$

No quantum-software development kit or circuit optimiser is used. Reversible basis transformations are compiled as permutations; local channels are applied by Kraus contraction; reduced states are obtained by explicit partial traces. The four history labels encode two logical bits, and the allowed predicate is parity,

$$p = (0, 1, 1, 0). \quad (32)$$

Thus each predicate fibre contains two histories and can hide one bit.

The default history first adds B into W modulo four and copies W into M . In round r , after memory formation in the gate sequence, a configured challenge parameter c_r is applied. The simulator does not model an independent challenge source, conceal c_r from earlier gates, or score the ZR-3 random-access game. The transducer computes

$$d_r = m_{r \bmod 2} \oplus c_r \quad (33)$$

into G , conditionally flips a world bit, and uncomputes G . The predicate oracle reads the final internal state:

$$p_{\text{int}} = \text{parity}(W) \oplus \bigoplus_r d_r = \text{parity}(B) \quad (34)$$

on the intended path. It never reads B directly. The phase ancilla starts in $|-\rangle$, so a controlled bit flip writes $(-1)^{p_{\text{int}}}$. The history is then inverted and $H \otimes H$ maps the parity phase pattern to $|11\rangle_B$.

The metrics are normalised multipath l_1 coherence

$$V_{l_1} = \frac{1}{K-1} \sum_{b \neq b'} |(\rho_B)_{bb'}|, \quad K = 4, \quad (35)$$

reset population

$$F_{\text{reset}} = \Pr[W = 0, M = 0, G = 0], \quad (36)$$

predicate fidelity $F_p = \Pr[B = 3]$ after readout, and conditional Holevo information $\chi(H : WMG|P)$. The first quantity is a normalised multipath coherence measure; it is not interchangeable with every two-path fringe convention.

6.2 Ideal cases and mandatory controls

Table 4: Exact density-matrix results. Numerical residues below 10^{-14} are printed as zero. The last column is the sum of separately evaluated controlled-record and environmental Holevo quantities. It is a bookkeeping sum of marginals, not a bound on a correlated joint record.

Case	V_{l_1}	F_{reset}	F_p	marginal sum (bits)
Ideal coherent	1.000	1.000	1.000	0.000
Classical mixture	0.000	1.000	0.250	0.000
Memory retained	0.000	0.250	0.250	1.000
Direct phase	1.000	1.000	1.000	0.000
Action bypassed	1.000	1.000	0.000	0.000
Environment overlap 0.5	0.500	1.000	0.625	0.811
Inverse failure 0.03	0.826	0.874	0.869	0.114

Figure 1 confirms the control logic. The classical mixture resets but carries no phase-sensitive coherence. Retaining memory leaves one bit within each parity fibre and removes interference. The directly compiled phase matches the full ideal endpoint exactly, numerically illustrating the endpoint theorem. Bypassing the internal action preserves coherence and reset but sends the phase to the wrong parity output; this is a white-box check that the trusted oracle depends on the declared trajectory, not a device-independent witness.

6.3 Environmental records

The environment model assigns pure records with constant pairwise overlap η . Tracing out the environment multiplies every off-diagonal branch block by η . The simulation reproduces the analytic laws to floating-point precision:

$$V_{l_1} = \eta, \quad F_p = \frac{1}{4} + \frac{3}{4}\eta, \quad \chi(H : E|P) = h_2\left(\frac{1+\eta}{2}\right). \quad (37)$$

At $\eta = 0.5$, the corresponding values are 0.5, 0.625, and 0.811278 bits. Reset remains exactly one throughout the sweep, directly demonstrating that local reset is not recoherence.

Four-history parity protocol and controls

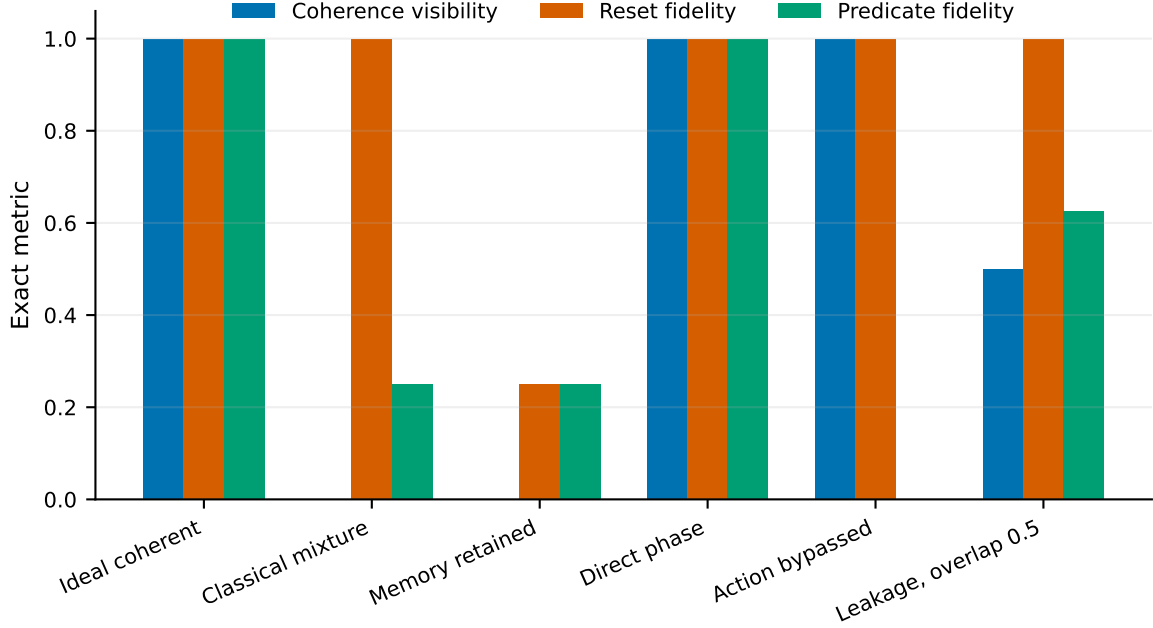


Figure 1: Exact metrics for the four-history protocol and controls. The direct-phase implementation deliberately equals the full trusted history at the endpoints. The action-bypassed circuit keeps the same internal predicate oracle but removes its causal action, causing the intended readout to fail.

6.4 Comparative Markovian channels

After each addressed logical operation, the simulator can apply qudit dephasing, full qudit depolarisation, or relaxation of every excited level toward zero. These are explicit comparative channels, not calibrations for a physical processor. In particular, four-level relaxation is not identical to independent amplitude damping on two hardware qubits.

Figure 3 shows that the metrics respond differently. Computational-basis dephasing leaves reset populations unchanged while reducing coherence and predicate fidelity. Depolarisation reduces all three. Relaxation can favour the zero reset subspace while still degrading the logical phase. This is why a single scalar fidelity cannot characterise the protocol.

6.5 Inverse reliability and depth

Imperfect inversion is modelled as a stochastic skipped inverse:

$$\mathcal{E}_{\text{inv}}(\rho) = (1 - \epsilon)U^\dagger \rho U + \epsilon \rho. \quad (38)$$

At $\epsilon = 0.03$ per inverse operation, visibility is 0.825798, reset fidelity 0.874200, predicate fidelity 0.869349, and the controlled records retain 0.113596 bits within parity classes. The model is intentionally simple and must not be confused with coherent over-rotation or leakage outside the logical Hilbert space.

The depth sweep increases challenge-responsive rounds from one to eight, corresponding to 11 through 53 logical operations, with per-addressed-operation dephasing 0.003 and depolarisation 0.001. Every ideal circuit remains exact. At 53 operations the noisy simulation gives

$$V_l = 0.892778, \quad F_{\text{reset}} = 0.969516, \quad F_p = 0.919584. \quad (39)$$

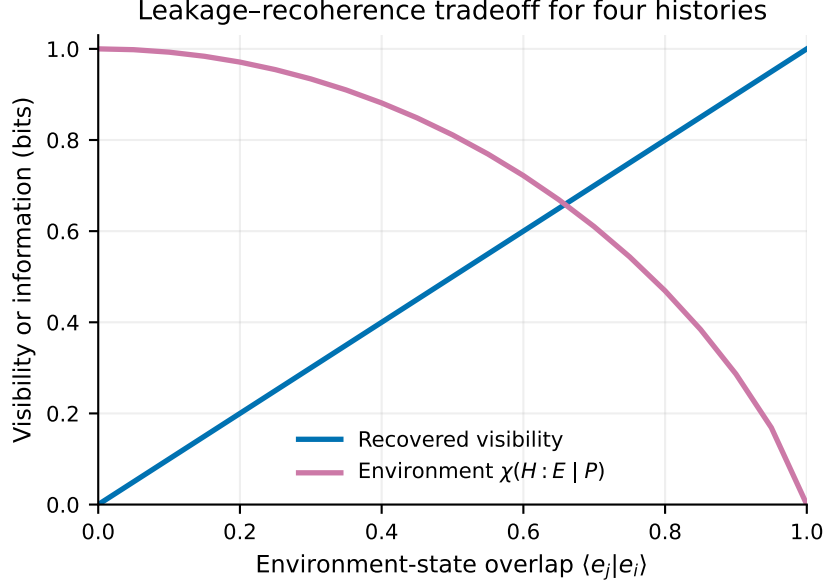


Figure 2: Exact constant-overlap environmental model. Increasing record overlap removes distinguishability inside each parity fibre and restores control coherence. The curve is a controlled analytic example, not a universal law for structured environments.

The numerical curve quantifies this model only; it is not a universal observer-complexity law.

6.6 Numerical claims and their boundary

Fourteen automated tests check permutation bijectivity, density-matrix invariants, trace preservation, Kraus completeness, exact ideal reset, classical-mixture loss, retained-record leakage, direct-phase equivalence, action dependence, the analytic overlap law, monotonic selected noise sweeps, and byte-for-byte reproducibility of generated tables under a fixed seed. These tests validate the implemented artifact. They do not test an independent late-challenge source, establish that a black-box device executed an irreducible history, certify joint leakage from correlated marginal records, or provide a quantum computational advantage.

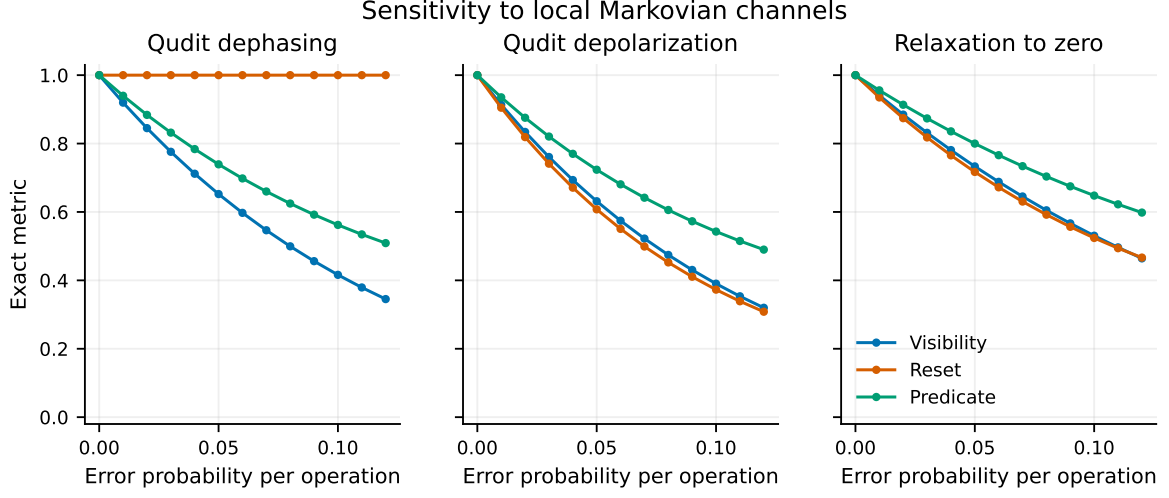
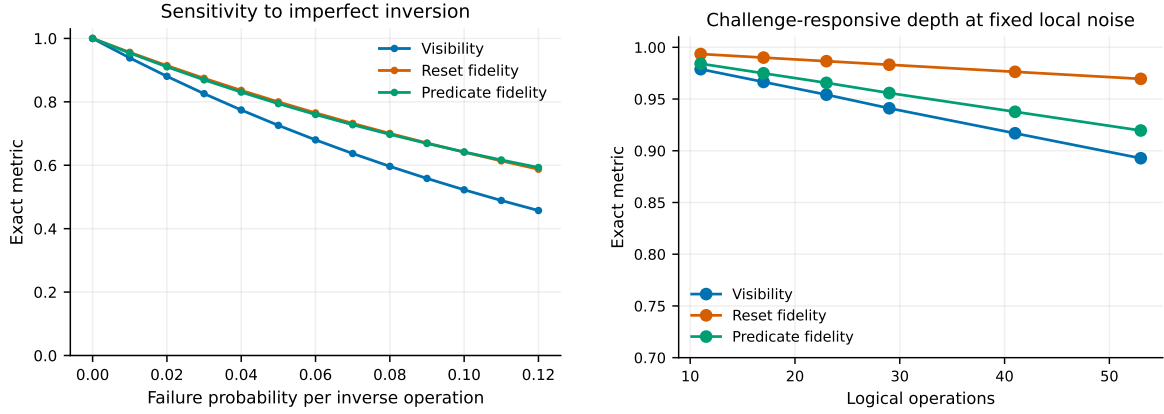


Figure 3: Sensitivity to three local Markovian channels applied after each addressed logical operation. Channel definitions are given in Appendix B; the horizontal parameter is not a reported hardware error rate.



(a) Stochastic skipped inverse.

(b) Fixed local noise and increasing rounds.

Figure 4: Practical recovery sensitivity. Every point is an exact density-matrix result for the channel model; finite-shot samples are stored separately in the generated data.

7 Experimental feasibility and scaling

7.1 What can be implemented now

ZR-0 and ZR-1 require only a path/control qubit, a memory qubit, a phase operation, and optionally one environmental marker. They are ordinary small coherent circuits and can be implemented with superconducting qubits, trapped ions, neutral-atom registers, or photonic path-polarisation degrees of freedom. Their scientific value is calibration: they test phase conventions, record accounting, and the difference between reset and recoherence.

ZR-2 and ZR-3 require coherent memory across several intervention times, conditional reversible gates, and an inverse whose error is low enough to expose the final phase. They remain small logical experiments, but the appropriate platform depends on the question. Trapped ions offer long-lived internal states and flexible connectivity at comparatively slow gate rates. Superconducting processors offer fast gates and mature calibration but require careful routing and suffer shorter coherence. Photonic implementations naturally expose path information and erasure but persistent multistep memory and deterministic interactions are harder. The present artifact is hardware-independent and makes no hardware execution claim.

7.2 Resource accounting

Let D_H be the forward-history depth and D_p the clean predicate depth. A literal inverse implementation has depth

$$D_{\text{total}} \geq 2D_H + D_p + D_{\text{interfere}}, \quad (40)$$

before routing, error correction, or verification overhead. If an exposed coherence is multiplied by λ_j at interval j , then

$$V \leq V_0 \prod_{j=1}^N \lambda_j. \quad (41)$$

For identical intervals, $V \leq V_0 \lambda^N \simeq V_0 e^{-N(1-\lambda)}$ near $\lambda = 1$. The forward-inverse structure therefore turns modest per-step losses into an exponential practical scaling, even though no special fundamental “observer complexity” term appears in unitary mechanics.

The simulation reports qubits, channels, depth proxy, and all noise parameters. It does not convert those counts into a universal energy cost. Ideal uncomputation is logically reversible and cannot be assigned $k_B T \ln 2$ per cleaned bit as though it were blind irreversible erasure [50, 1]. Finite-time control, refrigerator overhead, active reset, syndrome extraction, and discarded records can dissipate work, but their costs depend on the physical implementation. Recent complexity-constrained thermodynamics makes this distinction especially important [40, 41].

7.3 A staged experimental route

1. **Calibration:** implement ZR-0, direct-phase equality, no-predicate, and retained-memory controls.
2. **Record overlap:** implement a tunable marker and compare measured visibility with independently tomographed marker fidelity.
3. **Trusted causal agent:** implement the multistep reversible transducer and ablate observation, memory update, and action gates one at a time.
4. **Late challenge:** randomise the challenge only after memory formation; enforce physical isolation between the response module and the history label during the response window.
5. **Targeted process witness:** reconstruct only the temporal correlators needed to exclude the chosen bounded-memory class rather than performing full process-tensor tomography.

6. **Logical scaling:** compare visibility and reset with error-detected or error-corrected implementations at matched logical depth.

Steps 1–3 are feasible small-circuit demonstrations. Step 4 is feasible under trusted timing and connectivity assumptions. Step 5 is the first point at which the project could make a genuinely causal certification claim, and its witness has not yet been derived. Scaling to macroscopic natural records is not on this route: those records have already proliferated into uncontrolled environments outside the inverse.

7.4 Interpreting anomalous loss

Reduced dynamics are routinely nonunitary because of an unmodelled environment. A visibility deficit must first be tested against preparation and measurement error, coherent miscalibration, leakage levels, crosstalk, drift, relaxation, dephasing, loss, imperfect inversion, and an incomplete environmental model. Only a residual with preregistered controls, platform replication, and quantitatively excluded conventional channels could constrain a fundamental nonunitary model. Even then it would constrain a parameterised model; it would not by itself select an interpretation of quantum mechanics.

8 Discussion and conclusions

The central positive result is modest but exact: a quantum process may create distinguishable internal records, compute a property from them, convert that property to phase, and uncompute every controlled record while retaining an interferometric effect. Internal processing and conditional action do not introduce a new fundamental source of decoherence. What matters is whether history-dependent information remains outside the inverse and whether accumulated control errors preserve the required phase.

The central negative result is equally important: the same endpoint channel can be implemented without the advertised internal history. Therefore, a final interference fringe cannot certify an agent, a memory depth, or a unique temporal narrative. The direct-phase control is supposed to agree. Historical authenticity becomes a scientifically distinct question only after an access model introduces intervention times, late inputs, locality, black-box dynamics, or resource bounds.

Zero record gives a useful accounting rule. It is not enough to set a named memory qubit to zero; every residual system must be checked for distinctions that the allowed predicate does not reveal. Conditional mutual information captures this within-predicate requirement, provided the predicate is non-injective. The four-history benchmark exists specifically to prevent a formally correct but vacuous privacy score.

The analytic late-challenge protocol moves one step beyond a static ancilla. A classical response at a later time depends on information acquired before the challenge, and a bounded classical-memory comparison gives a quantitative success limit. The coherent phase variant uses the same response function, but its endpoint task is not the classical random-access score, and the current simulator does not model challenge independence. Both statements remain architecture-relative. If a shortcut device can access the history label when the response is produced, it can insert the phase directly. A publishable causal advance requires a witness or interactive task that makes the forbidden access operational rather than rhetorical.

Three research directions are defensible. First, process-tensor and temporal-correlation tools may yield a targeted memory-dimension witness compatible with a later inverse. Second, quantum-algorithmic-measurement methods may establish a coherent-versus-incoherent query separation for a black-box history property. Third, simulator-based quantum zero-knowledge ideas may inspire a composable definition in which a verifier learns a predicate while any

allowed residual view is simulable from that predicate alone. None of these results is claimed here.

The programme should be discontinued as a distinct research framing if every candidate task compiles to a known phase oracle under the same access assumptions, if memory cannot be causally witnessed, or if no new resource bound or experimentally distinguishable prediction results. A rigorous negative result would still be valuable: it would identify exactly why reversible internal narrative adds no operational resource beyond coherent computation.

In its present form, the project is a completed baseline rather than a completed research field. It supplies definitions, no-go statements, reference circuits, controls, reproducible numerics, and a falsifiable next target. Its scientific claim is not that quantum mechanics exposes alternative universes. It is that the boundary between a record-rich internal process and its recoverable coherence can be formulated as a precise causal-information task, and that any stronger claim must survive endpoint equivalence, transcript accounting, and explicit comparison models.

A Additional formal details and proofs

A.1 Process-tensor representation

Endpoint channels suppress temporal information by construction. A process tensor retains it. For intervention times $t_0, \dots, t_{T-1}$, let $\mathcal{A}_{x_t}^{(t)}$ be the completely positive map chosen at time t , with classical or coherently controlled setting x_t . The joint outcome probability has the generalised Born-rule form

$$p(\mathbf{a} \mid \mathbf{x}) = \text{Tr} \left[\Upsilon_{T:0} \left(A_{a_{T-1}|x_{T-1}}^{(T-1)} \otimes \dots \otimes A_{a_0|x_0}^{(0)} \right)^\top \right], \quad (42)$$

where the A 's are Choi operators and the transpose convention depends on the chosen link product. The positive operator $\Upsilon_{T:0}$ obeys linear causality constraints ensuring that later choices do not signal to earlier outcomes [12, 11].

A closed circuit corresponds to inserting fixed operations into every slot. The contraction then produces one channel $\mathcal{E}_{T:0}$. Many process tensors and many Stinespring realisations have the same contraction. Consequently, claims about internal memory require a family of interventions rich enough to exclude the selected null class.

One operational memory measure is the minimum dimension of a system passed between adjacent slots in a realisation compatible with the observed process. This dimension is interface-relative: moving a degree of freedom across the declared system–environment boundary changes the count. Dimension witnesses based on temporal correlations avoid full reconstruction but certify only relative to their assumptions.

A.2 Endpoint theorem for interactive strategies

The endpoint theorem extends to a verifier that exchanges messages at several times if the dishonest class may reproduce the entire honest comb.

Theorem A.1 (No unrestricted realisation soundness). *Let an honest implementation, including every conditional response to verifier messages, define an admissible quantum comb $\mathfrak{H}$. If the dishonest comparison class contains any physical realisation of the same comb and is not constrained to use the designated internal registers or causal decomposition, then no verifier interacting only through the declared ports has a positive completeness–soundness gap for the statement that the designated internal realisation occurred.*

Proof. The probability of every verifier transcript is obtained by linking the verifier strategy to the Choi operator of $\mathfrak{H}$. A second Stinespring realisation of the same comb has the same Choi operator at all ports, hence the same transcript distribution and acceptance probability. If the honest implementation is accepted with probability c , this alternative is accepted with probability c , forcing the soundness parameter $s \geq c$. $\square$

Nontrivial soundness therefore requires the comparison class to exclude the effective comb by a physical or computational restriction. Examples include spacelike or shielded interfaces, maximum communication dimension, bounded causal memory, delayed unpredictable inputs, query access to an unknown process, or a cryptographic assumption. Merely naming an internal wire does not restrict the class.

A.3 Visibility under independent records

For K alternatives,

$$|\Psi\rangle = \sum_{b=0}^{K-1} \alpha_b e^{i\phi_b} |b\rangle |r_b\rangle \quad (43)$$

gives

$$(\rho_B)_{bb'} = \alpha_b \alpha_{b'}^* e^{i(\phi_b - \phi_{b'})} \langle r_{b'} | r_b \rangle. \quad (44)$$

Every pairwise coherence is weighted by the corresponding record overlap. If step j writes into an independent environmental factor,

$$|r_b\rangle = \bigotimes_{j=1}^T |e_{b,j}\rangle, \quad (45)$$

then

$$\langle r_{b'} | r_b \rangle = \prod_{j=1}^T \langle e_{b',j} | e_{b,j} \rangle. \quad (46)$$

For two alternatives with per-step overlap magnitude γ_j ,

$$V_T = V_0 \prod_{j=1}^T \gamma_j. \quad (47)$$

If $\gamma_j = 1 - \ell_j$ and $\ell_j \ll 1$, then

$$-\log(V_T/V_0) = -\sum_j \log(1 - \ell_j) = \sum_j \ell_j + O\left(\sum_j \ell_j^2\right). \quad (48)$$

The product law is not universal: correlations between environmental factors or later reinteraction can produce revivals. Those effects require an explicit process-tensor model.

A.4 Mixed records and a joint recovery criterion

Let $f(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1$ denote root fidelity and $D(\rho, \sigma) = \|\rho - \sigma\|_1/2$. The Fuchs–van de Graaf inequalities [48] give

$$1 - f(\rho, \sigma) \leq D(\rho, \sigma) \leq \sqrt{1 - f(\rho, \sigma)^2}. \quad (49)$$

By Uhlmann’s theorem [47], the maximum two-path coherence compatible with mixed residual records ρ_R^0, ρ_R^1 is their root fidelity. An actual experiment can have lower visibility because its chosen purifications need not maximise their overlap. Therefore,

$$V \leq f(\rho_R^0, \rho_R^1), \quad D(\rho_R^0, \rho_R^1)^2 + V^2 \leq 1. \quad (50)$$

These are established information–disturbance consequences, not a project-specific observer law.

The correct sufficient recovery condition is joint. Suppose

$$F(\rho_{BR}, |\psi_\phi\rangle\langle\psi_\phi|_B \otimes \sigma_R) \geq 1 - \varepsilon, \quad |\psi_\phi\rangle = \frac{|0\rangle + e^{i\phi}|1\rangle}{\sqrt{2}}, \quad (51)$$

where F is squared fidelity and R includes every relevant record. Fuchs–van de Graaf bounds the trace distance by $\sqrt{\varepsilon}$. For

$$X_\phi = e^{-i\phi}|0\rangle\langle 1| + e^{i\phi}|1\rangle\langle 0|, \quad (52)$$

whose operator norm is one,

$$\text{Tr}[(X_\phi \otimes \mathbb{I})\rho_{BR}] \geq 1 - 2\sqrt{\varepsilon}. \quad (53)$$

A marginal statement about one memory register does not meet the premise.

A.5 Conditional leakage calculation

For a predicate value p , define

$$\bar{\rho}_R^p = \sum_{h:p(h)=p} q(h|p) \rho_R^h. \quad (54)$$

The within-fibre Holevo information is

$$\chi_p = S(\bar{\rho}_R^p) - \sum_{h:p(h)=p} q(h|p) S(\rho_R^h), \quad (55)$$

and

$$I(H : R|P) = \sum_p q_p \chi_p. \quad (56)$$

For pure residual states, $\chi_p = S(\bar{\rho}_R^p)$. In the ZR-4 parity example with a perfect retained two-bit transcript, $H(H|P) = 1$ bit and the transcript determines H , so $L_{\text{ZR}} = 1$ bit. Exact common-state reset gives zero. If the two states in each fibre have pure-state overlap γ and equal probability, the eigenvalues of their mean are $(1 \pm |\gamma|)/2$, giving

$$L_{\text{ZR}} = h_2\left(\frac{1 + |\gamma|}{2}\right), \quad (57)$$

where h_2 is binary entropy. This function falls continuously from one bit at $\gamma = 0$ to zero at $\gamma = 1$.

A.6 Late-challenge enumeration

For completeness, consider all deterministic one-bit encodings $m = f(h_0, h_1)$. Randomisation cannot improve a linear average-success objective beyond a deterministic extreme point. Each encoding partitions the four history strings into at most two cells. Given one cell and challenge x , the optimal decoder returns the majority value of coordinate x in that cell.

Up to relabelling, the nonconstant partitions have cell-size patterns $1 + 3$ or $2 + 2$. For a $1 + 3$ partition, the singleton contributes two correct challenge cases and the three-element cell contributes four, for six correct cases out of eight. For a $2 + 2$ partition:

- storing h_0 , h_1 , or their complements gives perfect response for one challenge and chance response for the other, again $6/8$;
- storing parity makes both coordinates balanced inside each cell and gives $4/8$.

Thus the optimal one-classical-bit value is $3/4$. A zero-bit response gives $1/2$, and a two-bit memory stores the full pair and gives one. If a qubit is allowed, encoding states around a great circle gives the 2-to-1 quantum random-access-code value $(1 + 1/\sqrt{2})/2$ [49], so the memory type is part of the null-class definition.

A.7 Elementary noise laws

The phase-flip channel

$$\mathcal{D}_\lambda(\rho) = (1 - \lambda)\rho + \lambda Z\rho Z \quad (58)$$

multiplies an off-diagonal element by $1 - 2\lambda$. After n independent applications,

$$V_n = V_0 |1 - 2\lambda|^n. \quad (59)$$

The convention matters: another common dephasing parameterisation multiplies coherence directly by $1 - \lambda$. The code documents which convention each curve uses.

Let the implemented inverse have coherent error

$$\tilde{U}_H^\dagger = U_H^\dagger e^{-i\epsilon K}. \quad (60)$$

For internal history state $|h_b\rangle$,

$$A_b = \langle h_b | e^{-i\epsilon K} | h_b \rangle \quad (61)$$

$$= 1 - i\epsilon \langle K \rangle_b - \frac{\epsilon^2}{2} \langle K^2 \rangle_b + O(\epsilon^3), \quad (62)$$

so

$$|A_b|^2 = 1 - \epsilon^2 \text{Var}_b(K) + O(\epsilon^3). \quad (63)$$

Coherent error can shift phase at first order while degrading return probability only at second order. Measuring one interferometric quadrature is therefore insufficient to distinguish contrast loss from phase rotation.

A.8 Proposed resource quantities

The following are research targets, not established monotones. For an allowed-control class $\mathfrak{U}$, define a recoherable causal complexity

$$\text{RCC}_{\mathfrak{U}}(v, \varepsilon) = \sup_{\Theta} \{ \log d_{\min}(\Theta) : V_{\text{excess}}(\Theta) \geq v, L_{\text{ZR}}(\Theta) \leq \varepsilon \}. \quad (64)$$

Without an interface, a dimension cap, or a control restriction, the quantity can be trivial or unbounded. A zero-record proof cost is better kept as a Pareto tuple

$$\text{C}_{\text{ZR}} = (N_q, D, N_a, N_{\text{shots}}, N_{\text{queries}}) \quad (65)$$

rather than collapsed with arbitrary weights. Finally, a coherent-access advantage should compare calls to the same process instance with coherent temporal control against independent preparations and classical intermediate records. Merely comparing a pure control with a dephased mixture is not a complexity separation.

B Simulation and reproduction details

B.1 Basis and gate construction

The subsystem order is B, W, M, G, A with dimensions 4, 4, 4, 2, 2. A basis index is mapped to a tuple by row-major tensor indexing. Every reversible classical transformation is evaluated on all 256 tuples and accepted only if the output indices form a permutation. Application to a density matrix uses

$$\rho'_{\pi(i), \pi(j)} = e^{i(\theta_i - \theta_j)} \rho_{ij}. \quad (66)$$

The inverse is another explicit permutation. Unit tests apply each forward–inverse pair on the whole basis, not just on the default initial state.

The initial coherent state is

$$|\Psi_0\rangle = \frac{1}{2} \sum_{b=0}^3 |b\rangle_B |0\rangle_W |0\rangle_M |0\rangle_G |-\rangle_A. \quad (67)$$

The incoherent control replaces the branch state by $\sum_b |b\rangle\langle b|/4$. The direct-phase control applies parity to the phase-kickback ancilla without instantiating W, M, G .

B.2 Kraus channels

The dephasing implementation uses

$$K_0 = \sqrt{1-p} \mathbb{I}_d, \quad K_j = \sqrt{p} |j\rangle\langle j|, \quad j = 0, \dots, d-1. \quad (68)$$

It multiplies local off-diagonal matrix elements by $1-p$ and preserves populations. The depolarising channel is

$$\mathcal{E}_{\text{dep}}(\rho) = (1-p)\rho + \frac{p}{d} \mathbb{I}_d \otimes \text{Tr}_{\text{target}} \rho. \quad (69)$$

Its Kraus set contains $\sqrt{1-p} \mathbb{I}_d$ and $\sqrt{p/d} |i\rangle\langle j|$ for all i, j .

The qudit relaxation channel uses

$$K_0 = |0\rangle\langle 0| + \sqrt{1-p} \sum_{j=1}^{d-1} |j\rangle\langle j|, \quad (70)$$

$$K_j = \sqrt{p} |0\rangle\langle j|, \quad j = 1, \dots, d-1. \quad (71)$$

All three sets satisfy $\sum_j K_j^\dagger K_j = \mathbb{I}$, checked numerically.

B.3 Environmental Schur channel

Pure environmental records with Gram matrix

$$\Gamma_{bb'} = \begin{cases} 1, & b = b', \\ \eta, & b \neq b' \end{cases} \quad (72)$$

induce the Schur channel

$$\rho_{bb'} \mapsto \Gamma_{bb'} \rho_{bb'} \quad (73)$$

on the branch blocks. For $0 \leq \eta \leq 1$, the Gram matrix is positive semidefinite and the channel is completely positive. The complementary conditional information is computed by constructing a square root of Γ , treating its rows as a valid set of record-state vectors, and evaluating the within-parity Holevo information.

B.4 Generated artifacts

The command

```
python scripts/run_all.py
```

runs the tests and then regenerates:

- data/benchmark_summary.csv;
- data/noise_sweep.csv;
- data/environment_sweep.csv;
- data/inversion_sweep.csv;

- `data/depth_sweep.csv`;
- `data/reference_branch_states.npz`;
- `data/metadata.json`;
- every PDF and PNG figure used in the manuscript.

Exact density-matrix quantities are deterministic. Seed 20260812 controls only the illustrative 8192-shot multinomial samples; seed 20260813 controls generated challenge strings in the depth sweep. Metadata record Python and library versions, Hilbert-space dimensions, row counts, command line, seed, and noise defaults.

B.5 Numerical precision and omissions

Density matrices are symmetrised after the main protocol to remove floating-point anti-Hermitian residue. Probabilities are clipped only at the final scalar level within roundoff. Tests require unit trace and positive semidefiniteness to tolerance. No sampling uncertainty is attached to exact curves. The CSV benchmark includes finite-shot counts so that downstream inference code can be tested without confusing those counts with experimental evidence.

The simulator omits coherent calibration drift, correlated and non-Markovian noise, leakage out of the declared Hilbert space, readout error, pulse compilation, crosstalk, error mitigation, logical error correction, and thermodynamic work. Adding one of these effects requires a stated channel or Hamiltonian and a new claim-ledger row; it must not be hidden inside an empirical fitting parameter.

C Claim audit, failure modes, and reproducibility contract

C.1 Claim classes

Table 5: Status of the main claims in this artifact.

ID	Statement	Status	Evidence and boundary
K1	A clean predicate phase survives exact uncomputation.	Established	Standard reversible computation; exact algebra and unit tests.
K2	A perfect external transcript is incompatible with nonzero two-path visibility.	Established	Complementarity and pure-record theorem.
D1	A named-register reset does not imply recoherence.	Derived	Explicit orthogonal-environment counterexample.
D2	A known closed protocol is endpoint-equivalent to its effective unitary.	Derived	Equality of channels; direct-phase control.
D3	Conditional leakage is nonvacuous only when an allowed predicate has a multi-history fibre.	Derived	Conditional mutual information identity.
N1	The reference circuits realise the analytic ideal and control predictions.	Numerical	Deterministic exact simulation and automated tests.
N2	The reported noisy curves follow the declared channel models.	Numerical	Generated CSV files, metadata, and tests.

ID	Statement	Status	Evidence and boundary
H1	The conjunction of causal-memory certification, predicate-only disclosure, final-record decoupling, recoherence, and an explicit anti-shortcut access model appears underexplored.	Novelty hypothesis	A bounded scoping search found separate clauses but cannot prove absence or priority.
C1	A coherent-over-incoherent complexity separation exists for a zero-record task.	Conjecture	Motivated by QUALM results; not proved here.
O1	A joint device-independent causal-memory and recoherence witness can be built.	Open objective	No such witness is supplied in this version.

C.2 Failure and falsification matrix

Table 6: Observations that invalidate a protocol claim before any foundational interpretation is considered.

Observation	Consequence
The coherent preparation and classical mixture give the same phase-sensitive statistics.	No coherence-dependent effect has been established.
The direct-phase control differs from the trusted ideal closed circuit.	The implementation or analysis is inconsistent; it does not establish history authenticity.
Removing the memory-update or action gate leaves the advertised predicate unchanged.	The named agent component is not causally necessary for that predicate.
Work ancillas or an environment retain distinguishable within-predicate states.	The zero-record claim fails even if the visible memory reads zero.
Only postselected subensembles display fringes, but the report treats them as unconditional.	The recovery mechanism has been misidentified.
An injective predicate is used to report $I(H : R P) = 0$.	The privacy metric is vacuous.
The null class can access the label and insert the phase directly.	The causal soundness gap collapses.
The ideal simulator cannot invert the declared circuit.	The protocol definition is internally inconsistent.

C.3 Interpretive neutrality

The density operator, channels, process tensor, and Born probabilities in this manuscript can be described using several interpretations of quantum mechanics. Calling coherent components branches is compatible with a relative-state description but adds no exclusive prediction. A successful experiment would establish the operational clauses in its access model. It would not demonstrate communication with a naturally decohered macroscopic world, and it would not single out an ontology.

C.4 Reproducibility contract

The public artifact includes:

1. source code implementing every simulated channel and metric;
2. tests of unitary validity, trace preservation, positivity, analytic limiting cases, and mandatory controls;
3. a single command that regenerates data and figures;

4. deterministic seeds recorded in machine-readable metadata;
5. CSV outputs containing the plotted values rather than only raster figures;
6. dependency bounds and a clean-environment installation path;
7. this claim ledger and a dated novelty audit;
8. the LaTeX source, BibTeX library, compiled PDF, and visual-render checks.

Numerical agreement does not prove hardware feasibility beyond the specified noise channels. Likewise, the absence of a source in the scoping review does not prove priority. Both limitations are preserved in the repository rather than removed from the final narrative.

C.5 Successor map

The next minimum scientifically decisive result is one of:

1. derive a multitime inequality with an explicit bounded-memory or no-memory null class, implement it, and combine it with a matched recoherence run;
2. construct a black-box task and prove a coherent-access query separation that still obeys the zero-record condition;
3. prove an impossibility result showing that causal certification and final physical transcript decoupling cannot be combined under a stated access model;
4. reduce the proposed primitive completely to an existing process-witness, zero-knowledge, or channel-recovery theorem and abandon the distinct framing.

Any of these outcomes would be more informative than increasing circuit size without improving identifiability.

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